S7.3a Body

S7.3(A)

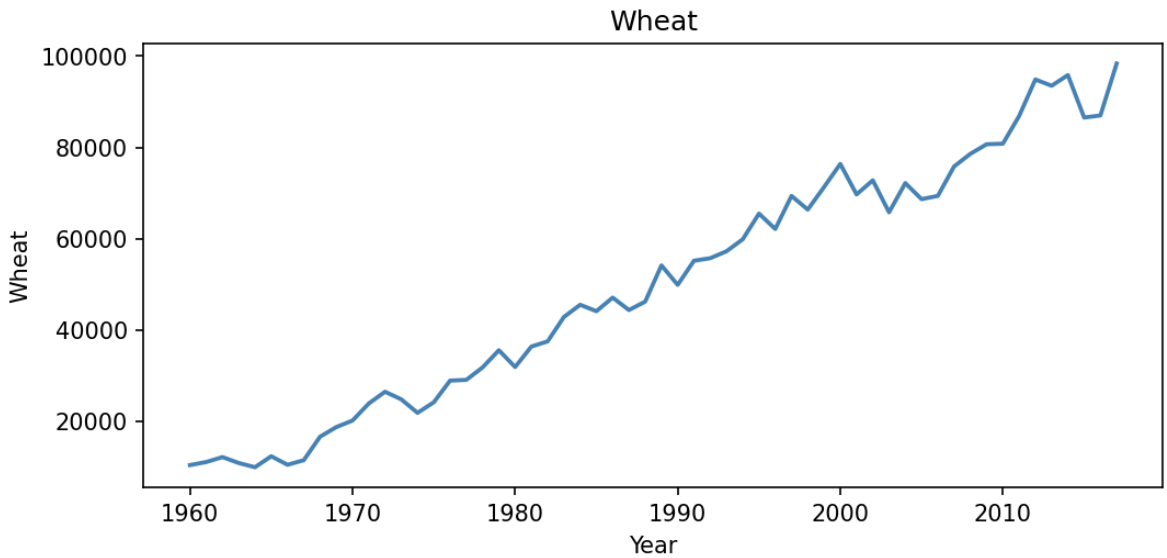

Figure 1: Time-series plot of wheat

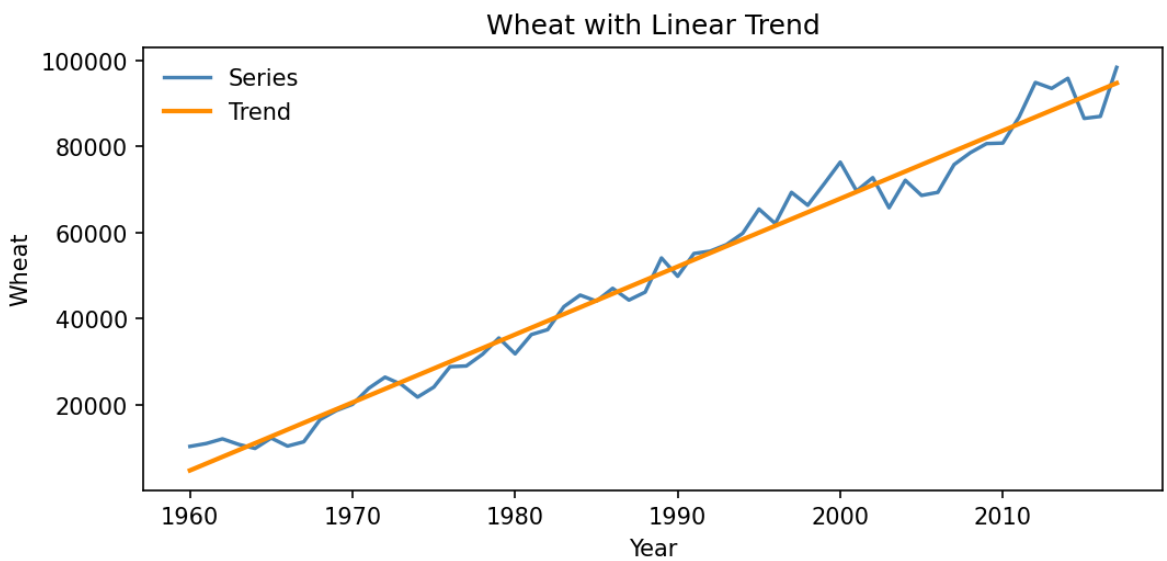

Figure 2: Wheat with a quadratic deterministic trend

The series shows a strong upward deterministic trend. A linear trend in time fits nearly as well as a quadratic one; AIC/BIC slightly favor the linear specification. A log-linear trend is also plausible if growth is proportional. Economically, the trend is consistent with technological improvements (Green Revolution), expanded irrigation, fertilizer use, improved seed varieties, and policy support that raised yields and production over time. (exponential growth) or a quadratic trend in time; the quadratic captures visible curvature in levels. Economically, the trend is consistent with technological improvements (Green Revolution), expanded irrigation, fertilizer use, improved seed varieties, and policy support that raised yields and production over time.

Method	RMSE	AIC	BIC	n
linear	3756.459	1123.420	1127.541	58
quadratic	3744.537	1125.051	1131.232	58
cubic	3744.701	1125.056	1131.237	58
log-linear	10548.084	-17.531	-13.411	58
log-quadratic	4225.527	-95.285	-89.103	58
hp	3516.936			58
moving_average	2345.683			58
lowess	3217.104			58

Table 1: Trend fit summary for wheat

S7.3b Body

S7.3(B)

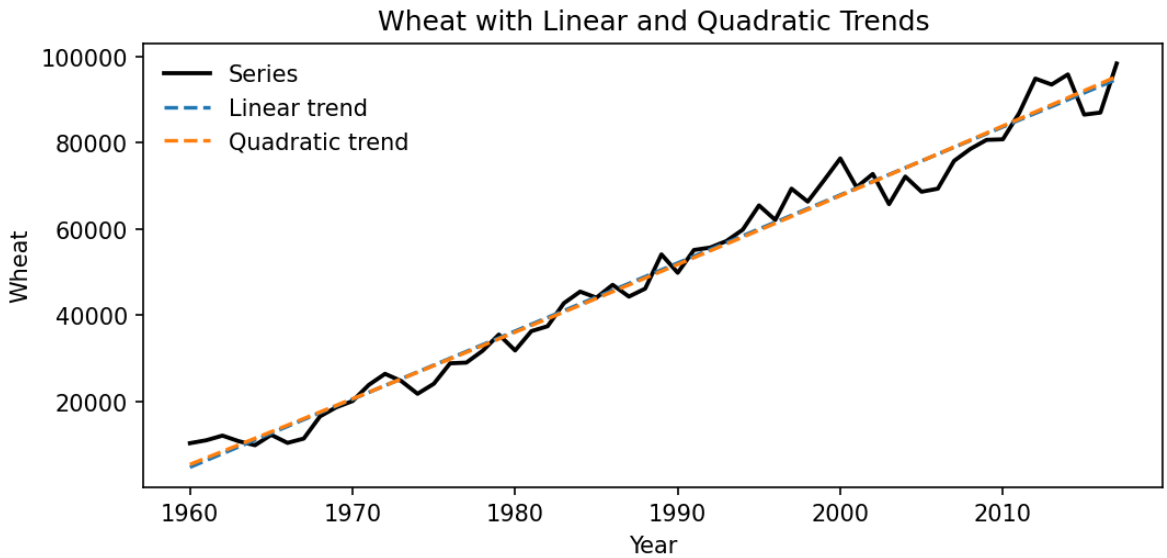

Figure 3: Wheat series with linear and quadratic trend fits (dashed)

AIC: linear=1123.420, quadratic=1125.051; BIC: linear=1127.541, quadratic=1131.232. AIC and BIC both prefer the linear trend. This makes intuitive sense because the quadratic term is small and adds little visible curvature; information criteria penalize the extra parameter when the fit gain is minimal.

S7.3c Body

S7.3(C)

Using the AIC-preferred linear trend, the 2018 forecast for wheat production is $\hat{y}_{2018} = 96309.4$ with a 95% confidence interval [94272.0, 98346.9]. This forecast assumes the deterministic trend is correctly specified, the error term is mean-zero with constant variance, and there are no structural breaks or policy/technology shocks between the sample and 2018.

S7.3d Body

S7.3(D)

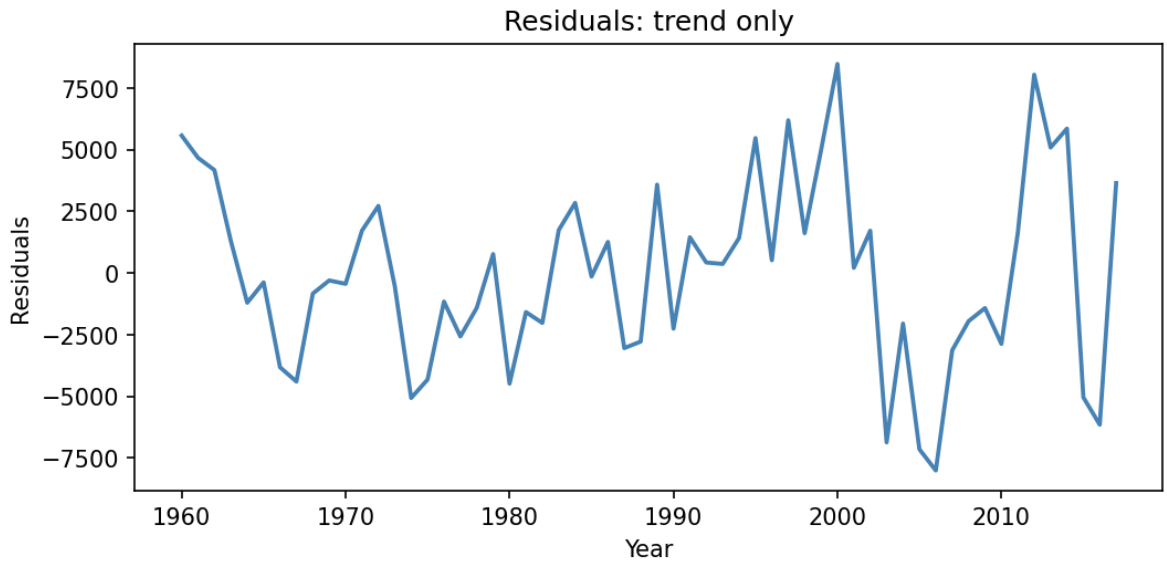

Figure 4: Residuals from trend-only model

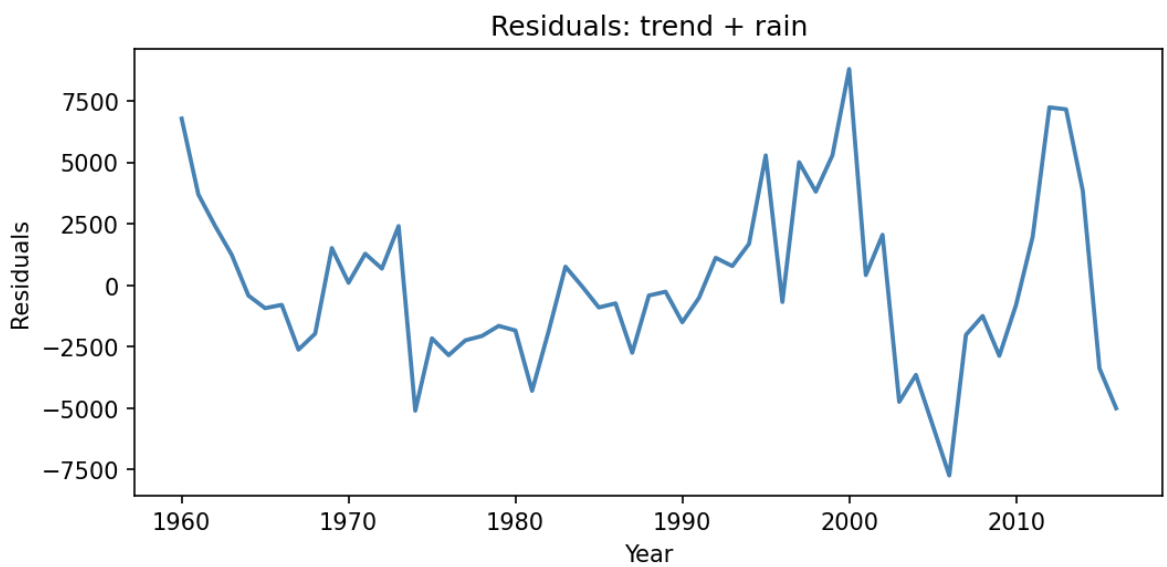

Figure 5: Residuals from trend + rain model

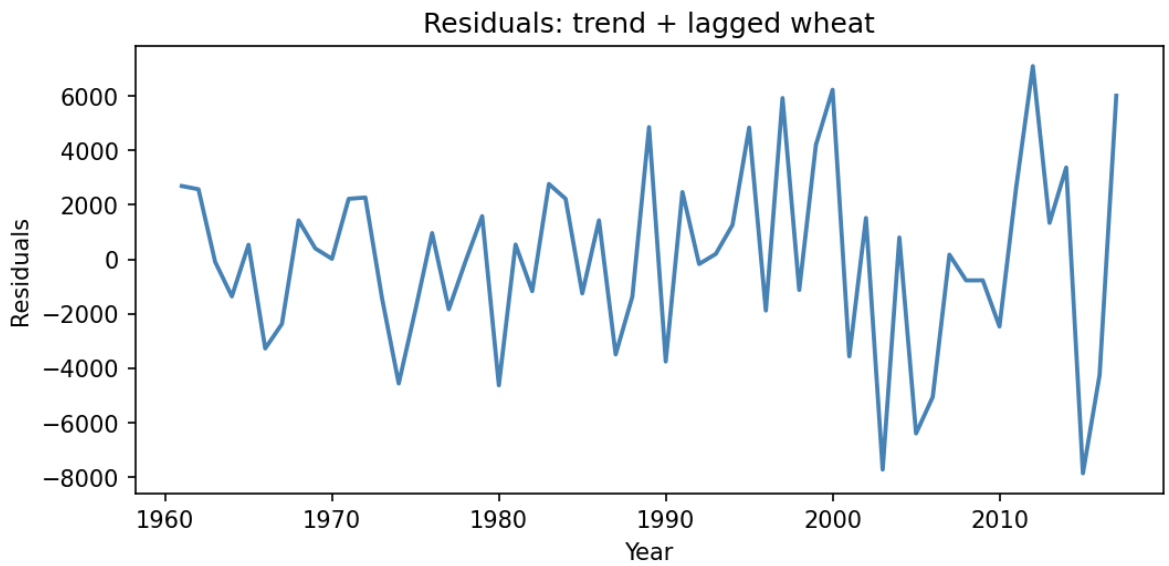

Figure 6: Residuals from trend + lagged wheat model

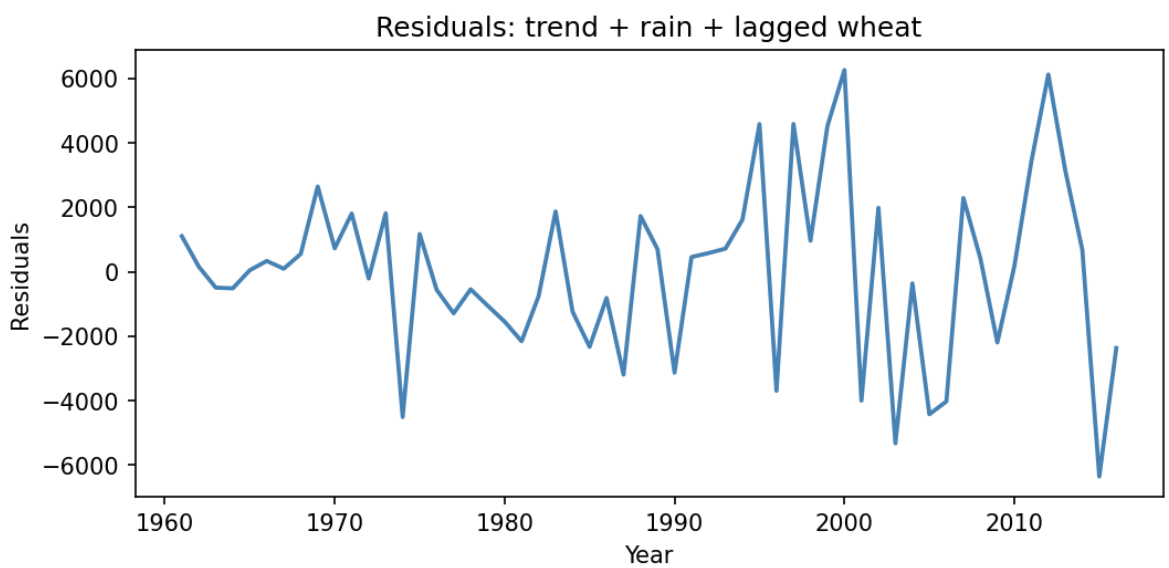

Figure 7: Residuals from trend + rain + lagged wheat model

Trend-only: $DW = 1.093$; Ljung–Box $Q(4) = 16.136$ ($p = 0.003$). Trend + rain: $DW = 0.766$; Ljung–Box $Q(4) = 27.315$ ($p = 0.000$). Trend + lagged wheat: $DW = 2.035$; Ljung–Box $Q(4) = 2.738$ ($p = 0.603$). Trend + rain + lagged wheat: $DW = 1.840$; Ljung–Box $Q(4) = 3.665$ ($p = 0.453$). Interpretation: adding lagged wheat substantially reduces serial correlation relative to trend-only or trend+rain, suggesting persistence in output that is not explained by rainfall alone. Rain may help explain levels but does not eliminate autocorrelation without a dynamic term.

S7.3e Body

S7.3(E)

We augment the trend model with rainfall to see whether it reduces residual dependence. Lag selection for Granger tests uses AIC with maxlags=4; selected lag = 1.

Series	Test	Stat	p	df	Dist	Reject
wheat (level)	ADF	0.216	0.973	4	ADF	No
wheat (level)	KPSS	1.249	0.010	4	KPSS	Yes
wheat (level)	PP	0.934	0.994	11	PP	No
wheat (diff)	ADF	-5.541	0.000	3	ADF	Yes
wheat (diff)	KPSS	0.106	0.100	5	KPSS	No
wheat (diff)	PP	-13.712	0.000	11	PP	Yes
rain (level)	ADF	0.115	0.967	7	ADF	No
rain (level)	KPSS	0.981	0.010	3	KPSS	Yes
rain (level)	PP	-8.703	0.000	11	PP	Yes
rain (diff)	ADF	-3.199	0.020	8	ADF	Yes
rain (diff)	KPSS	0.167	0.100	15	KPSS	No
rain (diff)	PP	-37.913	0.000	11	PP	Yes

Table 2: Stationarity tests for wheat and rain in levels and first differences

Test	Stat	p	df	Dist	Reject
Granger: rain $\rightarrow$ wheat	16.010	0.000	1, 53	F	Yes
Granger: wheat $\rightarrow$ rain	21.956	0.000	1, 53	F	Yes
Lead exogeneity (rain)	0.129	0.721	1, 50	F	No

Table 3: Exogeneity and Granger causality tests

Trend + rain model residual diagnostics: $DW = 0.766$; Ljung–Box $Q(4) = 27.315$ with $p = 0.000$. If residual serial correlation persists, the inference and forecast intervals from (c) are understated; a dynamic specification (e.g., AR errors, adding lagged wheat, or differencing) would be more appropriate.